

R Data Practices — Quiz

Functions and Concepts · 20 Questions

True/False and multiple choice · One correct answer per question

Topics covered:

`filter()` · `arrange()` · `select()` · `mutate()` · `group_by()` + `summarise()` · `pipe |>` · `srvyr weights` · `here()`

Data Practices 2026 — Functions & Concepts

Instructions

1. Fill in your name, student ID, and today's date in the fields below.
2. Click the radio button next to your chosen answer for each question.
3. Each question has exactly one correct answer.
4. Save the completed PDF and submit it to the course portal.

Student information

Full name: _____

Student ID: _____

Date: _____

1

Q1. Sofia has a spreadsheet of all her cats. She wants to keep only rows where age > 5. Which function does she need?

- ☐ A. `arrange()`
- ☐ B. `select()`
- ☐ C. `filter()`
- ☐ D. `mutate()`

2

Q2. `filter(name == "Luna")` keeps rows where the name column contains the text Luna.

- ☐ True
- ☐ False

3 **Q3.** Tomás filters a pizza menu for vegetarian AND gluten-free options. He writes: `filter(vegetarian == TRUE, gluten_free == TRUE)`. What does the comma do?

- ☐ A. Separates two separate `filter()` calls
- ☐ B. Acts as OR — rows matching either condition are kept
- ☐ C. Acts as AND — only rows matching both conditions are kept
- ☐ D. Causes an error — you cannot have two conditions

4 **Q4.** Nadia has a table of her favourite songs with a column `play_count`. She wants the most-played song first. Which code is correct?

- ☐ A. `filter(desc(play_count))`
- ☐ B. `arrange(play_count)`
- ☐ C. `arrange(desc(play_count))`
- ☐ D. `sort(play_count, decreasing = TRUE)`

5 **Q5.** After calling `arrange()`, the data frame has fewer rows than before.

- ☐ True
- ☐ False

6 **Q6.** Kenji has a dog adoption table with 12 columns. He only wants `dog_name`, `breed`, and `age`. Which function should he use?

- ☐ A. `filter()`
- ☐ B. `select()`
- ☐ C. `slice()`
- ☐ D. `keep()`

7 **Q7.** Amara types: `select(full_name = name, city)`. What happens to the column previously called "name"?

- ☐ A. It is deleted from the data frame
- ☐ B. It is kept with its original name
- ☐ C. It is renamed to `full_name`
- ☐ D. An error is thrown

8

Q8. Beatriz has a cake recipe table with a column grams. She wants a new column kg equal to grams / 1000. Which code is correct?

- ☐ A. `mutate(grams / 1000 -> kg)`
- ☐ B. `mutate(kg == grams / 1000)`
- ☐ C. `mutate(kg = grams / 1000)`
- ☐ D. `add_column(kg = grams / 1000)`

9

Q9. `mutate()` removes the columns it used as inputs so only the new column remains.

- ☐ True
- ☐ False

10

Q10. Lior has a table of hiking trails. He wants the 5 trails with the longest distance. Which function gives exactly those rows?

- ☐ A. `head(5)`
- ☐ B. `filter(rank == 1:5)`
- ☐ C. `slice_max(distance, n = 5)`
- ☐ D. `top(distance, 5)`

11

Q11. Priya has 500 restaurant reviews with columns city and rating. She writes: `group_by(city) |> summarise(avg = mean(rating))`. How many rows does the result have?

- ☐ A. 500 — one per review
- ☐ B. 1 — the overall mean
- ☐ C. One row per unique city
- ☐ D. It depends on the value of rating

12

Q12. Adding `na.rm = TRUE` inside `mean()` tells R to skip missing values instead of returning NA.

- ☐ True
- ☐ False

13 **Q13.** After `group_by(species) |> summarise(total = n())`, Marco's result is still grouped. Which function removes the grouping?

- ☐ A. `degroup()`
- ☐ B. `ungroup()`
- ☐ C. `reset_groups()`
- ☐ D. `group_by(NULL)`

14 **Q14.** Yuki reads: `dogs |> filter(breed == "Shiba") |> arrange(age)`. What does `|>` mean?

- ☐ A. "or" — keep rows matching either condition
- ☐ B. "and then" — pass the result to the next function
- ☐ C. Divide the left side by the right side
- ☐ D. Assign the result to a new variable

15 **Q15.** In modern R (2021+), `|>` is the recommended native pipe operator.

- ☐ True
- ☐ False

16 **Q16.** Chiara downloaded a national health survey with a column `survey_weight`. To get a nationally representative mean she should:

- ☐ A. Use `mean(bmi)` directly — weights don't matter for means
- ☐ B. Create `as_survey_design(weights = survey_weight)` then use `survey_mean()`
- ☐ C. Multiply every value by its weight before taking the mean
- ☐ D. Use `weighted_mean()` from base R

17 **Q17.** Calling `mean()` directly on a column from a weighted survey sample gives the same result as a properly weighted estimate.

- ☐ True
- ☐ False

18

Q18. Inside a `srvyr summarise()` call, which function computes a weighted proportion from a 0/1 binary variable?

- ☐ A. `mean()`
- ☐ B. `prop()`
- ☐ C. `survey_mean()`
- ☐ D. `wtd.mean()`

19

Q19. Oliver moves his R project from "C:/Users/oliver/Desktop/" to "D:/research/". His script uses `here("data", "birds.csv")`. After moving, the script:

- ☐ A. Breaks — `here()` hardcodes the original path
- ☐ B. Still works — `here()` builds the path relative to the `.Rproj` file
- ☐ C. Still works — R always searches the entire computer for files
- ☐ D. Throws a warning but loads the file anyway

20

Q20. Using `setwd()` inside a script is considered best practice for managing file paths in R projects.

- ☐ True
- ☐ False